Two Constructions of Rotation-Symmetric Bent Functions Outside the Completed Maiorana–McFarland Class with Any Possible Algebraic Degree

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Abstract

Rotation-symmetric Boolean functions form an important class of cryptographically significant Boolean functions. In 2017, Su and Tang proposed in [IEEE TIT 63(7): 4658–4667, 2017] an infinite class of rotation-symmetric bent functions of every possible algebraic degree. In this paper, we present two constructions of rotation-symmetric bent functions outside the completed Maiorana–McFarland class on $n = 30 \cdot 7^j$ variables with $j \geq 0$ and $n = 70t$ variables with $t \geq 1$, respectively. Each of the two constructions generates bent functions of every possible algebraic degree ranging from 3 to $n/2$. Since the algebraic degree of an n-variable bent function is at most $n/2$ and every quadratic bent function belongs to the completed Maiorana–McFarland class, the interval from 3 to $n/2$ is the full possible degree range for bent functions outside this class. To the best of our knowledge, these are the first infinite constructions of rotation-symmetric bent functions in which functions have algebraic degrees ranging from 3 to $n/2$ while remaining entirely outside the completed Maiorana–McFarland class.

Keywords. Boolean function; Rotation-symmetric bent function; Frobenius automorphism; algebraic degree; completed Maiorana–McFarland class.

1 Introduction

Boolean bent functions were introduced by Rothaus in 1976 [18]. Among the well-known constructions are the Maiorana–McFarland class and the partial-spread class introduced by McFarland and Dillon, respectively [11, 7]. These constructions and their extensions have played a central role in the theory of bent functions; see [3, 12, 1] for more details.

The Boolean functions that are invariant under the action of a cyclic shift on the input variables are called rotation-symmetric functions [15]. This class is of particular interest because its functions can be stored in less space and it contains Boolean functions with desirable cryptographic properties, such as high nonlinearity and high algebraic degree [19]. In particular, this class contains bent functions, which achieve the maximum possible nonlinearity in an even number of variables. Quadratic and cubic rotation-symmetric bent functions were studied in [6, 8], and later constructions produced functions of high algebraic degree [2, 24, 25, 20, 21, 5]. In particular, Su and Tang constructed rotation-symmetric bent functions of any possible algebraic degree, and Tang et al. subsequently gave finite field constructions of bent idempotents with the corresponding algebraic degrees [22, 23].

It is well-known that $\mathbb{F}_{2^n}$ has a normal basis over $\mathbb{F}_2$ for every positive integer n [13]. With respect to such a basis, the Frobenius automorphism $\alpha \mapsto \alpha^2$ over $\mathbb{F}_{2^n}$ induces a cyclic shift of

the coordinates. Consequently, a Boolean function on $\mathbb{F}_{2^n}$ satisfies $f(\alpha^2) = f(\alpha)$ if and only if its coordinate representation is rotation-symmetric [2, 23]. A Boolean function satisfying this equality is called an idempotent function. Accordingly, the finite field representation of a rotation-symmetric bent function is called a bent idempotent. We use this equivalence to represent rotation-symmetric Boolean functions over finite fields.

The completed Maiorana–McFarland class $\mathcal{M}^{\#}$ consists of the bent functions that are extended-affine equivalent to members of the Maiorana–McFarland class. The algebraic degree alone does not determine whether a bent function belongs to $\mathcal{M}^{\#}$. Cubic bent functions outside $\mathcal{M}^{\#}$ were constructed in [17, 10], and further infinite families were obtained in [14, 9]. Rotation-symmetric bent functions outside $\mathcal{M}^{\#}$ were investigated in [16]; among other results, that work gives a family of functions of algebraic degree $n/2$. However, these results do not give, for each fixed number n of variables in an infinite parameter set, functions outside $\mathcal{M}^{\#}$ with any possible algebraic degree ranging from 3 to $n/2$.

In this paper, we present two constructions that fill this gap. The first applies to $n = 30 \cdot 7^j$ for every $j \geq 0$ and views $\mathbb{F}_{2^{30 \cdot 7^j}}$ as a degree three extension of $\mathbb{F}_{2^{10 \cdot 7^j}}$, while the second applies to $n = 70t$ for every $t \geq 1$ and uses the two reciprocal cubic factors of $z^7 - 1$. In both constructions, reciprocal polynomial factors determine pairs of Frobenius-invariant subspaces U and W such that the absolute-trace pairing $(\alpha, \beta) \mapsto \mathrm{Tr}_1^n(\alpha\beta)$ is nondegenerate on $U \times W$. This property allows summation over W to reduce each Walsh coefficient to a Walsh coefficient of a bent function on the remaining variables, thereby proving bentness. For every n in either parameter set, the corresponding construction yields functions of every algebraic degree from 3 to $n/2$. To prove that these functions lie outside $\mathcal{M}^{\#}$, we use M-subspaces [7, 17, 9]; an M-subspace is a linear subspace on which the second-order derivative $D_a D_b f$ of a Boolean function f is identically zero for all a and b in that subspace. We show that every M-subspace has dimension at most $14 \cdot 7^j < 15 \cdot 7^j = n/2$ in the first construction and at most $34t < 35t = n/2$ in the second. Hence none of the constructed functions belongs to $\mathcal{M}^{\#}$. Since every quadratic bent function belongs to $\mathcal{M}^{\#}$ and the algebraic degree of an n-variable bent function is at most $n/2$, the interval from 3 to $n/2$ is the full possible degree range outside this class. To the best of our knowledge, these are the first infinite constructions of rotation-symmetric bent functions in which functions have algebraic degrees ranging from 3 to $n/2$ while remaining entirely outside the completed Maiorana–McFarland class.

The remainder of this paper is organized as follows. Section 2 introduces the notation and preliminary results. Section 3 establishes the results used in both constructions. The constructions for $n = 30 \cdot 7^j$ and $n = 70t$ are given in Sections 4 and 5, respectively. Section 6 compares the resulting functions with existing constructions, and Section 7 concludes the paper.

2 Preliminaries

In this section, we introduce the notation and recall the results used throughout the paper. We first review Boolean functions, algebraic degree, rotation symmetry, and finite field representations. We then introduce bent functions, extended-affine equivalence, and M-subspaces.

Let $\mathbb{F}_2 = \{0, 1\}$ be the field with two elements, and let $\mathbb{F}_2^n$ denote the vector space of binary n-tuples. An n-variable Boolean function is a map $f : \mathbb{F}_2^n \to \mathbb{F}_2$, and we use $\mathcal{B}_n$ to denote the

set of all such functions. The basic representation of f is its truth table

$$f = [f(0,0,\ldots,0), f(1,0,\ldots,0), f(0,1,\ldots,0), \ldots, f(1,1,\ldots,1)].$$

In this paper, all additions of field elements and Boolean functions are over $\mathbb{F}_2$ and are denoted by $+$. We use $\oplus$ only for a direct sum of vector spaces; when no confusion is possible, a direct sum decomposition may also be written with $+$. For a subset A of a set, $\mathbf{1}_A$ denotes the indicator function of A. Let $\rho : \mathbb{F}_2^n \to \mathbb{F}_2^n$ be the cyclic-shift operator defined by $(x_0, x_1, \ldots, x_{n-1}) \mapsto (x_{n-1}, x_0, \ldots, x_{n-2})$. For $j \geq 0$, let ρ^j denote the j-fold composition of ρ with itself, where ρ^0 is the identity map on $\mathbb{F}_2^n$. Then $\rho^j : \mathbb{F}_2^n \to \mathbb{F}_2^n$ is given by $(x_0, x_1, \ldots, x_{n-1}) \mapsto (x_{-j}, x_{1-j}, \ldots, x_{n-1-j})$, where all subscripts are computed modulo n. A Boolean function $f \in \mathcal{B}_n$ is rotation-symmetric if $f(\rho(x)) = f(x)$ for every $x \in \mathbb{F}_2^n$ [15, 19]. Equivalently, $f(\rho^j(x)) = f(x)$ for every $x \in \mathbb{F}_2^n$ and every $j \geq 0$. It is known that every $f \in \mathcal{B}_n$ has a unique algebraic normal form (ANF)

$$f(x_1, \ldots, x_n) = \sum_{u \in \mathbb{F}_2^n} a_u \prod_{i=1}^{n} x_i^{u_i}, \quad a_u \in \mathbb{F}_2.$$

For $u = (u_1, \ldots, u_n) \in \mathbb{F}_2^n$, its Hamming weight is defined by $\mathrm{wt}(u) = \big|\{i : u_i = 1\}\big| = \sum_{i=1}^n u_i$. The algebraic degree of f, denoted by $\deg f$, is $\deg f = \max\{\mathrm{wt}(u) : a_u \neq 0\}$. Constant functions have algebraic degree zero, and functions of algebraic degree at most one are affine. It is well-known that an invertible affine change of variables preserves the algebraic degree of a Boolean function and if p and q are nonzero Boolean functions in disjoint sets of variables then $\deg(pq) = \deg p + \deg q$.

Let $\mathbb{F}_{2^n}$ be the finite field with 2^n elements. A basis of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ identifies $\mathbb{F}_{2^n}$ with $\mathbb{F}_2^n$. Every Boolean function on $\mathbb{F}_{2^n}$ has a unique univariate representation $f(\alpha) = \sum_{i=0}^{2^n-1} a_i \alpha^i$ of degree at most $2^n - 1$ as a polynomial function, where $a_i \in \mathbb{F}_{2^n}$ and $f(\alpha)^2 = f(\alpha)$ for every $\alpha \in \mathbb{F}_{2^n}$. If $i = \sum_{s=0}^{n-1} i_s 2^s$, put $\mathrm{wt}(i) = \sum_{s=0}^{n-1} i_s$. The algebraic degree of f in this presentation is $\max\{\mathrm{wt}(i) : a_i \neq 0\}$ [1]. A Boolean function $f : \mathbb{F}_{2^n} \to \mathbb{F}_2$ is called an idempotent if $f(\alpha^2) = f(\alpha)$ for every $\alpha \in \mathbb{F}_{2^n}$ [2, 23]. It should be noted that "idempotent" is the customary term for this invariance condition on a Boolean function; it is distinct from multiplicative idempotence in a ring. We also call such a function Frobenius-invariant in the sequel. The normal basis theorem guarantees the existence of an element $\theta \in \mathbb{F}_{2^n}$ such that $(\theta, \theta^2, \ldots, \theta^{2^{n-1}})$ is a basis over $\mathbb{F}_2$ [13]. Squaring cyclically permutes the coordinates in this basis. Therefore, idempotent Boolean functions and rotation-symmetric Boolean functions are the same functions under a normal basis. If $m \mid n$, the trace function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^m}$ is defined as

$$\mathrm{Tr}_m^n(\alpha) = \sum_{i=0}^{n/m-1} \alpha^{2^{mi}}.$$

We call Tr_m^n the trace function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^m}$ and write Tr_1^n for the trace function to $\mathbb{F}_2$.

All vector-space dimensions and algebraic degrees below are over $\mathbb{F}_2$ unless another base field is specified. For $f \in \mathcal{B}_n$, its Walsh transform is

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)+a\cdot x}, \quad a \in \mathbb{F}_2^n.$$

If one vector is represented with respect to a basis of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ and the other with respect to its trace-dual basis, their standard inner product is $\mathrm{Tr}_1^n(\alpha\beta)$. If $n = 2r$, then f is bent

if $|W_f(a)| = 2^r$ for every a. The algebraic degree of a bent function in $2r$ variables is at most r when $r \geq 2$ [18, 1]. For $a \in \mathbb{F}_2^n$, the first-order derivative of a Boolean function $f : \mathbb{F}_2^n \to \mathbb{F}_2$ at a vector a is defined by $D_a f(x) = f(x+a) + f(x)$. For $a, b \in \mathbb{F}_2^n$, the second-order derivative of f in the vectors a and b is defined by $D_a D_b f(x) = D_a(D_b f)(x) = f(x) + f(x+a) + f(x+b) + f(x+a+b)$. In particular, $D_a D_b f = D_b D_a f$.

The Maiorana–McFarland class $\mathcal{M}$ consists of the bent functions on $\mathbb{F}_2^r \times \mathbb{F}_2^r$ of the form $(x, y) \mapsto x \cdot \pi(y) + h(y)$, where π is a permutation of $\mathbb{F}_2^r$ and h is an arbitrary Boolean function on r variables [11, 7, 1]. Its completion under extended-affine equivalence is denoted by $\mathcal{M}^\#$. More precisely, two Boolean functions f and g on $\mathbb{F}_2^n$ are extended-affine equivalent if $g(x) = f(Lx + a) + \ell(x)$ for an invertible linear map L, a vector a, and an affine Boolean function ℓ. The class $\mathcal{M}^\#$ consists of all functions that are extended-affine equivalent to functions in $\mathcal{M}$. Let l, e be positive integers and $m = l + e$. For any $g \in \mathcal{B}_l$ and $h \in \mathcal{B}_e$, the Boolean function $f \in \mathcal{B}_m$ defined by $f(x_1, \ldots, x_m) = g(x_1, \ldots, x_l) + h(x_{l+1}, \ldots, x_m)$ is called the direct sum of g and h. This construction provides a simple way to obtain Boolean functions in more variables from functions in fewer variables; in particular, the direct sum of two bent functions is bent.

To prove that the constructed bent functions lie outside the completed Maiorana–McFarland class, we introduce the following definition.

Definition 2.1 ([17, 9])**.** Let $f : \mathbb{F}_2^n \to \mathbb{F}_2$ be a Boolean function. A linear subspace $Z \subseteq \mathbb{F}_2^n$ is called an M-subspace of f if

$$D_a D_b f = 0 \quad \text{for all } a, b \in Z.$$

The linearity index of f is

$$\operatorname{ind}(f) = \max\{\dim Z : Z \text{ is an } M\text{-subspace of } f\}.$$

A linear subspace $Z \subseteq \mathbb{F}_2^n$ is called a relaxed M-subspace of f if $D_a D_b f$ is constant for all $a, b \in Z$. The relaxed linearity index of f is

$$\text{r-ind}(f) = \max\{\dim Z : Z \text{ is a relaxed } M\text{-subspace of } f\}.$$

Every M-subspace is a relaxed M-subspace. Consequently,

$$\operatorname{ind}(f) \leq \text{r-ind}(f).$$

The reverse inequality does not hold in general. A bent function in $2r$ variables belongs to $\mathcal{M}^\#$ if and only if it has an r-dimensional M-subspace [17, 9, 16]. We shall use the necessary direction of this criterion.

Lemma 2.2 (see, e.g., [1])**.** *Every quadratic bent function in $2r$ variables belongs to $\mathcal{M}^\#$ [1].*

Lemma 2.2 shows that the algebraic degree ranges outside $\mathcal{M}^\#$ begin with algebraic degree three. Bent functions are neither constant nor affine, every quadratic bent function is already in $\mathcal{M}^\#$, and an n-variable bent function has algebraic degree at most $n/2$. We therefore consider exactly the degrees $3, 4, \ldots, n/2$ in our constructions below.

3 Basic Results Used in Both Constructions

In this section, we establish the results used in both constructions. We first combine a nondegenerate bilinear form with bent functions and derive an upper bound on the dimensions of the resulting M-subspaces. We then consider direct sums of a 10-variable cubic bent function and construct bent functions whose sums have prescribed algebraic degrees. Finally, we recall the form of Carlet's indirect sum construction needed below. Sections 4 and 5 apply these results to two different finite field decompositions.

3.1 Bent functions obtained from a nondegenerate bilinear form

Let U and W be binary vector spaces of the same dimension. A bilinear map $b : U \times W \to \mathbb{F}_2$ is nondegenerate if, for every nonzero $c \in U$, there is a $d \in W$ such that $b(c, d) = 1$, and the same condition holds after interchanging U and W. Equivalently, the map $c \mapsto b(c, \cdot)$ is an isomorphism from U to the dual space W^*. In this situation, we also say that b is nondegenerate on $U \times W$. Viewed as a Boolean function on $U \oplus W$, the map $q_b(u, w) = b(u, w)$ is a quadratic bent function. Indeed, after choosing bases of U and W, the nondegeneracy of b makes q_b linearly equivalent to the standard quadratic bent function $(u, w) \mapsto u \cdot w$.

The following theorem is used in both constructions. Its Walsh-transform calculation is the standard argument underlying the Maiorana–McFarland construction [11, 7, 1]. We also prove the upper bound on the dimensions of the M-subspaces, which will be used later in the proofs of both main theorems.

Theorem 3.1. *Let r and k be positive integers. Let V_0, U, W be binary vector spaces, and write $V = V_0 \oplus U \oplus W$, where the dimensions satisfy $\dim(V_0) = 2r$ and $\dim(U) = \dim(W) = k$. Let $b : U \times W \to \mathbb{F}_2$ be a nondegenerate bilinear map. Let $g_0, g_1 : V_0 \to \mathbb{F}_2$ be bent functions and $\eta : U \to \mathbb{F}_2$ be a Boolean function. For $x_{V_0} \in V_0$, $x_U \in U$, and $x_W \in W$, we define $h_\eta, h_{\eta,g_1} : V \to \mathbb{F}_2$ by*

$$h_\eta(x_{V_0}, x_U, x_W) = b(x_U, x_W) + g_0(x_{V_0}) + \eta(x_U),$$
$$\text{and} \qquad h_{\eta,g_1}(x_{V_0}, x_U, x_W) = b(x_U, x_W) + g_0(x_{V_0}) + \eta(x_U)\big(g_0(x_{V_0}) + g_1(x_{V_0})\big).$$

Thus V has the even dimension $2r + 2k$. Both functions are bent. If a linear automorphism of V preserves the three summands, the bilinear map b, g_0, and η, then h_η is invariant under this automorphism. If it also preserves g_1, then h_{η,g_1} is invariant under the same automorphism. Moreover,

$$\operatorname{ind}(h_\eta) \le k + \operatorname{ind}(g_0).$$

If η is nonconstant, then

$$\operatorname{ind}(h_{\eta,g_1}) \le k + \operatorname{ind}(g_0).$$

Proof. We first prove bentness and invariance. We first relate the two functions to standard bent-function constructions. We choose bases of U and W in which $b(x_U, x_W) = x_U \cdot x_W$. The function $q_b(x_U, x_W) + \eta(x_U)$ belongs to the Maiorana–McFarland class, and h_η is its direct sum with g_0. If we put $q_0 = q_b$ and $q_1 = q_b + \eta$, then both q_0 and q_1 are bent and $h_{\eta,g_1} = g_0 + q_0 + (g_0 + g_1)(q_0 + q_1)$. Thus h_{η,g_1} is precisely Carlet's indirect sum applied to g_0, g_1, q_0, q_1 (see Proposition 3.4 and [4]). These descriptions already explain the bentness assertions. For completeness, we give a direct Walsh-transform proof of bentness and invariance, and then prove the required bounds. For a binary vector space X, let X^* denote the dual

space of all $\mathbb{F}_2$-linear maps from X to $\mathbb{F}_2$. We take $(\alpha, \beta, \gamma) \in V_0^* \oplus U^* \oplus W^*$. This triple specifies the linear function $(x_{V_0}, x_U, x_W) \mapsto \alpha(x_{V_0}) + \beta(x_U) + \gamma(x_W)$ in the Walsh transform. We first sum over x_W. For each fixed $x_U \in U$, summation over $x_W \in W$ gives

$$\sum_{x_W \in W} (-1)^{b(x_U, x_W) + \gamma(x_W)}.$$

The nondegeneracy of b gives a unique $u_\gamma \in U$ such that $b(u_\gamma, x_W) = \gamma(x_W)$ for every $x_W \in W$. Therefore, we have

$$\sum_{x_W \in W} (-1)^{b(x_U, x_W) + \gamma(x_W)} = \begin{cases} 2^k, & x_U = u_\gamma, \\ 0, & x_U \neq u_\gamma. \end{cases}$$

Consequently,

$$W_{h_\eta}(\alpha, \beta, \gamma) = 2^k (-1)^{\eta(u_\gamma) + \beta(u_\gamma)} W_{g_0}(\alpha).$$

For h_{η, g_1}, the same summation over x_W gives

$$W_{h_{\eta,g_1}}(\alpha, \beta, \gamma) = 2^k (-1)^{\beta(u_\gamma)} \begin{cases} W_{g_0}(\alpha), & \eta(u_\gamma) = 0, \\ W_{g_1}(\alpha), & \eta(u_\gamma) = 1. \end{cases}$$

Since g_0 and g_1 are bent, every Walsh coefficient of h_{η,g_1} has magnitude $2^k \cdot 2^r = 2^{k+r}$. The invariance assertion follows directly from the hypotheses on the automorphism.

It remains to bound the dimensions of the M-subspaces. We take an M-subspace $Z \subseteq V$ of either h_η or h_{η,g_1}. We identify W with the subspace $\{0\} \oplus \{0\} \oplus W$ of V and put $Z_W = Z \cap W$. Let $p_U : V \to U$ be the canonical coordinate projection with respect to $V = V_0 \oplus U \oplus W$ and thus $p_U(x_{V_0}, x_U, x_W) = x_U$. Take $y = (y_{V_0}, y_U, y_W) \in Z$ and $d_W \in Z_W$. Under the above identification, d_W corresponds to the vector $(0, 0, d_W) \in Z$. The second-order derivative of the bilinear term with respect to y and $(0, 0, d_W)$ is $b(y_U, d_W)$, while every other term has zero second-order derivative involving $(0, 0, d_W)$. Since Z is an M-subspace, we have $b(y_U, d_W) = 0$.

We define the annihilator of Z_W in U by $Z_W^\perp = \{a_U \in U : b(a_U, d_W) = 0 \text{ for all } d_W \in Z_W\}$. Since $y \in Z$ and $d_W \in Z_W$ were arbitrary, the preceding equality gives $p_U(Z) \subseteq Z_W^\perp$. The bilinear map b is nondegenerate and $\dim U = \dim W = k$. Hence the map $\phi : U \to Z_W^*$ defined by $\phi(a_U)(d_W) = b(a_U, d_W)$ is surjective and has kernel $Z_W^\perp$. Indeed, ϕ is obtained by restricting to Z_W the isomorphism from U to W^* induced by b. Consequently,

$$\dim p_U(Z) \leq k - \dim Z_W.$$

We set $Z_0 = \ker(p_U|_Z)$ and let $p_{V_0} : V \to V_0$ be the canonical projection onto the V_0-component. For h_η, the space $p_{V_0}(Z_0)$ is an M-subspace of g_0. Indeed, every vector in Z_0 has zero U-component, so the terms $b(x_U, x_W)$ and $\eta(x_U)$ have zero second-order derivative with respect to vectors in Z_0, while the W-components do not occur in g_0. For h_{η,g_1}, we take two vectors $a = (a_{V_0}, 0, a_W) \in Z_0$ and $c = (c_{V_0}, 0, c_W) \in Z_0$. Since Z is an M-subspace, the second-order derivative of h_{η,g_1} with respect to a and c is zero, and therefore

$$D_{a_{V_0}} D_{c_{V_0}} g_0(x_{V_0}) + \eta(x_U) D_{a_{V_0}} D_{c_{V_0}} (g_0 + g_1)(x_{V_0}) = 0.$$

This identity holds for every $x_{V_0} \in V_0$ and $x_U \in U$. If η is nonconstant, it takes both values in $\mathbb{F}_2$. Choosing x_U with $\eta(x_U) = 0$ gives $D_{a_{V_0}} D_{c_{V_0}} g_0 = 0$ identically; choosing x_U with $\eta(x_U) = 1$ then gives $D_{a_{V_0}} D_{c_{V_0}} g_1 = 0$ identically. Thus $p_{V_0}(Z_0)$ is again an M-subspace of g_0.

In both cases, the kernel of $p_{V_0}|_{Z_0}$ consists exactly of the vectors whose V_0- and U-components vanish, and hence equals Z_W. By the rank–nullity theorem, $\dim Z_0 = \dim p_{V_0}(Z_0) + \dim Z_W$, and therefore $\dim Z_0 \le \mathrm{ind}(g_0) + \dim Z_W$. Similarly, $\dim Z = \dim p_U(Z) + \dim Z_0$. Combining these equalities with $\dim p_U(Z) \le k - \dim Z_W$ gives $\dim Z \le k + \mathrm{ind}(g_0)$ in both cases. This completes the proof. □

3.2 A cubic bent function on $10t$ variables

We fix an integer $t \ge 1$ and choose a normal basis $(\theta, \theta^2, \ldots, \theta^{2^{10t-1}})$ of $\mathbb{F}_{2^{10t}}$ over $\mathbb{F}_2$. For every $\alpha \in \mathbb{F}_{2^{10t}}$, write

$$\alpha = \sum_{i=0}^{10t-1} x_i(\alpha)\theta^{2^i}, \quad \text{where} \quad x_i(\alpha) \in \mathbb{F}_2.$$

Thus $x_i(\alpha)$ denotes the ith coordinate of α with respect to this normal basis. For a positive integer m, let $z_0, \ldots, z_{m-1}$ be Boolean variables, and let ρ be the cyclic shift defined in Section 2, applied here to m variables. For a monomial $\mu = \prod_{i \in I} z_i$, where $I \subseteq \{0, 1, \ldots, m-1\}$, its cyclic orbit is defined by

$$\mathcal{O}_m(\mu) = \{\mu \circ \rho^j : 0 \le j < m\}.$$

The corresponding orbit sum is

$$\mathrm{orb}_m(\mu) = \sum_{\nu \in \mathcal{O}_m(\mu)} \nu.$$

Since $\mathcal{O}_m(\mu)$ is a set, each distinct cyclic shift of μ occurs exactly once in $\mathrm{orb}_m(\mu)$.

For $(z_0, \ldots, z_9) \in \mathbb{F}_2^{10}$, we define $g' : \mathbb{F}_2^{10} \to \mathbb{F}_2$ by

$$\begin{aligned} g'(z_0, \ldots, z_9) = {} & \mathrm{orb}_{10}(z_0 z_5) + \mathrm{orb}_{10}(z_0 z_1 z_4) + \mathrm{orb}_{10}(z_0 z_1 z_8) \\ & + \mathrm{orb}_{10}(z_0 z_2 z_4) + \mathrm{orb}_{10}(z_0 z_2 z_7). \end{aligned}$$

A direct computation with the help of Magma gives the Walsh spectrum $\{\pm 32\}$ and $\text{r-ind}(g') = 4$. Hence $g' \in \mathcal{B}_{10}$ is a rotation-symmetric cubic bent function outside $\mathcal{M}^{\#}$. For related 10-variable cubic bent functions outside $\mathcal{M}^{\#}$ and direct-sum constructions based on such functions, see [17, 14] for examples. For $\alpha \in \mathbb{F}_{2^{10t}}$ and $0 \le j < t$, we put $x^{(j)}(\alpha) = (x_j(\alpha), x_{j+t}(\alpha), \ldots, x_{j+9t}(\alpha)) \in \mathbb{F}_2^{10}$. We define $g : \mathbb{F}_{2^{10t}} \to \mathbb{F}_2$ by

$$g(\alpha) = \sum_{j=0}^{t-1} g'(x^{(j)}(\alpha)), \quad \alpha \in \mathbb{F}_{2^{10t}}. \tag{1}$$

Proposition 3.2. *The Boolean function g defined in (1) is Frobenius-invariant and bent on $\mathbb{F}_{2^{10t}}$ with algebraic degree three. Moreover,*

$$\text{r-ind}(g) \le 4t < 5t. \tag{2}$$

Proof. It should be noted that under the coordinate permutation $(x_0, \ldots, x_{10t-1}) \mapsto (x^{(0)}, x^{(1)}, \ldots, x^{(t-1)})$, the function in (1) is the direct sum of t copies of g' on disjoint 10-variable blocks. Therefore, its Walsh transform is the product of the Walsh transforms of the blocks. Each factor has magnitude 2^5, so every Walsh coefficient of g has magnitude 2^{5t} and hence g is bent. Its algebraic degree is three because the cubic terms of distinct blocks cannot cancel with respect to direct sum. Squaring permutes the t blocks cyclically and applies one cyclic shift inside the 10-variable block when the block index returns to zero. Hence $g(\alpha^2) = g(\alpha)$ for every $\alpha \in \mathbb{F}_{2^{10t}}$.

We take a relaxed M-subspace $z \subseteq \mathbb{F}_2^{10t}$ of g, and for $0 \le j < t$, we let $p_j : \mathbb{F}_2^{10t} \to \mathbb{F}_2^{10}$ denote the projection onto the jth 10-variable block. Given $c_j, d_j \in p_j(z)$, we choose $c, d \in z$ with $p_j(c) = c_j$ and $p_j(d) = d_j$. The linear part of $D_c D_d g$ is a sum of linear functions in disjoint blocks. Restricting this linear part to the jth block gives the linear part of $D_{c_j} D_{d_j} g'$, so it vanishes. Thus $p_j(z)$ is a relaxed M-subspace of g' and has dimension at most four. The map $z \to \bigoplus_j p_j(z)$ is injective because a vector whose projection onto every coordinate block is zero is the zero vector. Therefore, $\dim z \le \sum_j \dim p_j(z) \le 4t$. This completes the proof. □

3.3 Pairs of bent functions whose sums have prescribed algebraic degrees

Let $r = 5t$. For $\alpha \in \mathbb{F}_{2^{10t}}$ and $0 \le i < r$, we set $y_i(\alpha) = x_i(\alpha) + x_{i+r}(\alpha)$, and write $y(\alpha) = (y_0(\alpha), \dots, y_{r-1}(\alpha)) \in \mathbb{F}_2^r$. For $(z_0, \dots, z_{r-1}) \in \mathbb{F}_2^r$ and $3 \le e \le r$, we let $\gamma_{e,0} : \mathbb{F}_2^r \to \mathbb{F}_2$ be given by $\gamma_{e,0}(z) = \mathrm{orb}_r(z_0 z_1 \cdots z_{e-1})$, and put $\gamma_{3,1} : \mathbb{F}_2^r \to \mathbb{F}_2$, $\gamma_{3,1}(z) = \mathrm{orb}_r(z_0 z_1 z_3)$. For $e \ge 4$, we set $\gamma_e = \gamma_{e,0}$. For $e = 3$, we choose the first of $\gamma_{3,0}$ and $\gamma_{3,1}$ whose cubic homogeneous part, after substituting $z_i = x_i + x_{i+r}$ for $0 \le i < r$, differs from the cubic homogeneous part of g. The monomials $z_0 z_1 z_2$ and $z_0 z_1 z_3$ belong to distinct cyclic orbits because $r = 5t \ge 5$. Hence $\gamma_{3,0}$ and $\gamma_{3,1}$ are distinct. The linear map $(x_0, \dots, x_{2r-1}) \mapsto (x_0 + x_r, \dots, x_{r-1} + x_{2r-1})$ is surjective. Hence the substitution $z_i = x_i + x_{i+r}$ is injective on the Boolean functions in $z_0, \dots, z_{r-1}$. Therefore, the two substituted cubic forms remain distinct, and at most one of them can equal the cubic homogeneous part of g. Thus this choice is well defined. We define $g'_e : \mathbb{F}_{2^{10t}} \to \mathbb{F}_2$ by

$$g'_e(\alpha) = \sum_{i=0}^{r-1} x_i(\alpha) x_{i+r}(\alpha) + \gamma_e(y(\alpha)), \quad \alpha \in \mathbb{F}_{2^{10t}}. \tag{3}$$

Proposition 3.3. *For every $3 \le e \le 5t$, the Boolean function g'_e defined in (3) is Frobenius-invariant and bent on $\mathbb{F}_{2^{10t}}$ with algebraic degree e, and*

$$\deg(g + g'_e) = e. \tag{4}$$

Proof. The main point is that, under an invertible linear change of variables, the function in (3) becomes a Maiorana–McFarland function. We give this change explicitly because it also determines the algebraic degree. Squaring cyclically shifts both the x_i and the y_i, so g'_e is Frobenius-invariant. Let us consider the linear change from $(x_0, \dots, x_{2r-1})$ to $(p_0, \dots, p_{r-1}, y_0, \dots, y_{r-1})$ defined by $p_i = x_i$ and $y_i = x_i + x_{i+r}$. Its inverse is $x_i = p_i$ and $x_{i+r} = p_i + y_i$, so the change is invertible over $\mathbb{F}_2$. Under this change of variables, (3) becomes

$$\sum_{i=0}^{r-1} p_i(y_i + 1) + \gamma_e(y).$$

This is a Maiorana–McFarland bent function with the permutation $(y_0, \dots, y_{r-1}) \mapsto (y_0 + 1, \dots, y_{r-1} + 1)$. By definition, orb_r is the sum of the distinct monomials in a cyclic orbit. Every monomial in $\mathrm{orb}_r(z_0 z_1 \cdots z_{e-1})$ has algebraic degree e, and the distinct monomials cannot cancel. Hence γ_e has algebraic degree e; this also covers $e = r$, when the orbit of $y_0 y_1 \cdots y_{r-1}$ has one element. This change of variables is invertible and therefore preserves algebraic degree. It follows that g'_e has algebraic degree e. If $e \ge 4$, its homogeneous part of algebraic degree e cannot cancel with the cubic function g. If $e = 3$, the choice of γ_3 prevents cancellation of the cubic homogeneous parts: $\gamma_{3,0}$ and $\gamma_{3,1}$ are the orbit sums corresponding to two distinct cyclic orbits and hence have different cubic parts. Thus we get (4). This completes the proof. □

The following indirect sum construction was introduced by Carlet [4]. We recall the form that will be used later in our constructions.

Proposition 3.4 ([4])**.** *Let f_0, f_1 be bent functions on an even-dimensional binary space V_1, and let q_0, q_1 be bent functions on an even-dimensional binary space V_2. Then the function*

$$h(x,y) = f_0(x) + q_0(y) + (f_0(x) + f_1(x))(q_0(y) + q_1(y)), \text{ where } (x,y) \in V_1 \oplus V_2,$$

is bent on $V_1 \oplus V_2$. Let L_1 and L_2 be linear automorphisms of V_1 and V_2, respectively. If f_0 and f_1 are invariant under L_1, and q_0 and q_1 are invariant under L_2, then h is invariant under $L_1 \oplus L_2$. Moreover, $h(x,y) + f_0(x) + q_0(y) = (f_0(x) + f_1(x))(q_0(y) + q_1(y))$. If $d_1 = \deg(f_0 + f_1) > 0$ and $d_2 = \deg(q_0 + q_1) > 0$, then $\deg(h + f_0 + q_0) = d_1 + d_2$.

This identity follows directly from the definition of h. The degree formula follows because its two factors depend on disjoint sets of variables.

4 Rotation-Symmetric Bent Functions Outside $\mathcal{M}^\#$ with Any Possible Algebraic Degree for $n = 30 \cdot 7^j$ Variables

In this section, for every $j \geq 0$ and every integer d satisfying $3 \leq d \leq n/2$, we construct an idempotent bent function of algebraic degree d outside $\mathcal{M}^\#$ on $\mathbb{F}_{2^n}$, where $n = 30 \cdot 7^j$. To carry out the construction, we put $t = 7^j$, $m = 10t$, and $n = 3m = 30t$, and view $\mathbb{F}_{2^n}$ as a degree three extension of $\mathbb{F}_{2^m}$. We decompose the kernel of the trace map Tr_m^n into Frobenius-invariant primary subspaces determined by the irreducible factors of $z^{2m} + z^m + 1$. One primary subspace corresponds to the self-reciprocal factor $z^2 + z + 1$, while the remaining primary subspaces occur in pairs corresponding to reciprocal irreducible factors. This decomposition provides the nondegenerate bilinear pairing and the vector-space decomposition required to apply Theorem 3.1.

4.1 Decomposition of the kernel of the trace function

We define $\sigma : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ by $\sigma(\alpha) = \alpha^2$ for $\alpha \in \mathbb{F}_{2^n}$, and we put $\tau = \sigma^m$. We also put $s = \mathbb{F}_{2^m}$ and $v = \ker \mathrm{Tr}_m^n$. Since $\mathbb{F}_{2^n}$ has degree three over $\mathbb{F}_{2^m}$, $\mathrm{Tr}_m^n(\alpha) = \alpha$ for $\alpha \in s$. Hence the trace function is surjective and $\mathbb{F}_{2^n} = s \oplus v$. Moreover, for $\alpha \in \mathbb{F}_{2^n}$ and $\beta \in s$, trace transitivity gives $\mathrm{Tr}_1^n(\alpha\beta) = \mathrm{Tr}_1^m(\beta\,\mathrm{Tr}_m^n(\alpha))$. The bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^m(\alpha\beta)$ on s is nondegenerate. With respect to the bilinear form on $\mathbb{F}_{2^n}$, we define $s^\perp = \{\alpha \in \mathbb{F}_{2^n} : \mathrm{Tr}_1^n(\alpha\beta) = 0 \text{ for every } \beta \in s\}$. The preceding identity proves $v = s^\perp$. Thus this direct-sum decomposition is also orthogonal.

In normal-basis coordinates, $\mathbb{F}_{2^n}$ is a cyclic module for σ and is represented by $\mathbb{F}_2[z]/(z^n - 1)$, where multiplication by z represents σ. Under this isomorphism, $\mathrm{Tr}_m^n = 1 + \sigma^m + \sigma^{2m}$ is multiplication by $1 + z^m + z^{2m}$. Since $z^n - 1 = (z^m - 1)(z^{2m} + z^m + 1)$ and the two factors are coprime, the Chinese remainder theorem gives two cyclic submodules. The trace operator is the identity on the first and zero on the second. Consequently, the kernel of Tr_m^n is precisely the cyclic submodule associated with $z^{2m} + z^m + 1 = (z^{10t} + z^{5t} + 1)^2$. Let $\overline{\mathbb{F}}_2$ denote the algebraic closure of $\mathbb{F}_2$. We choose a primitive root of unity $\zeta \in \overline{\mathbb{F}}_2$ of order $15t$. The roots of $z^{10t} + z^{5t} + 1$ are the elements ζ^b with $0 \leq b < 15t$ and $b \not\equiv 0 \pmod 3$. Their possible orders are of the forms $3 \cdot 7^i$ and $15 \cdot 7^i$ with $0 \leq i \leq j$.

Table 1: Irreducible factors of $z^{10t} + z^{5t} + 1$ over $\mathbb{F}_2$

Order of a root	Factor degree	Number of factors
3	2	1
15	4	2
$3 \cdot 7^i$, $1 \le i \le j$	$6 \cdot 7^{i-1}$	2
$15 \cdot 7^i$, $1 \le i \le j$	$12 \cdot 7^{i-1}$	4

For coprime positive integers a and d, we let $\operatorname{ord}_d(a)$ denote the multiplicative order of a modulo d, and we let φ denote Euler's totient function. We use $\operatorname{ord}_{7^i}(2) = 3 \cdot 7^{i-1}$ for $i \ge 1$. This identity follows by induction from $2^3 = 1 + 7$ and the binomial expansion of $(1 + c7^i)^7$ modulo 7^{i+2} for an integer c with $7 \nmid c$. Since $\operatorname{ord}_3(2) = 2$ and $\operatorname{ord}_5(2) = 4$, the Chinese remainder theorem (CRT) gives $\operatorname{ord}_{3 \cdot 7^i}(2) = 6 \cdot 7^{i-1}$ and $\operatorname{ord}_{15 \cdot 7^i}(2) = 12 \cdot 7^{i-1}$ for $i \ge 1$. The irreducible factors whose roots have order d with degree $\operatorname{ord}_d(2)$, and their number is $\varphi(d)/\operatorname{ord}_d(2)$. The resulting data are listed in Table 1.

For a monic polynomial $p \in \mathbb{F}_2[z]$ with nonzero constant term, we write $p^*(z) = z^{\deg p} p(z^{-1})$ for its reciprocal polynomial. An irreducible factor having a root of order d is self-reciprocal if and only if $2^k \equiv -1 \pmod{d}$ for some k. The factor $z^2 + z + 1$ associated with order three is self-reciprocal, whereas all the other factors occur in distinct reciprocal pairs. For order $3 \cdot 7^i$ with $i \ge 1$, a power of two cannot be -1 modulo 7^i because $\operatorname{ord}_{7^i}(2)$ is odd. For order $15 \cdot 7^i$, the congruence modulo three would require an odd exponent, whereas the congruence modulo five would require an exponent congruent to two modulo four. These two congruence conditions cannot hold simultaneously.

For every reciprocal pair, we choose the factor whose binary cyclotomic coset has the smaller least positive representative. We let $c \subseteq v$ be the primary component associated with $(z^2+z+1)^2$, let $u \subseteq v$ be the direct sum of the primary components associated with the chosen factors, and let $w \subseteq v$ be the direct sum of the components associated with their reciprocals. Here the p^2-primary component is the direct-sum component in the CRT that is isomorphic to $\mathbb{F}_2[z]/(p^2)$. Equivalently, it is the maximal σ-invariant subspace of kernel $\ker \operatorname{Tr}_m^n$ annihilated by $p(\sigma)^2$.

Proposition 4.1. *The spaces s, c, u, w above give a direct-sum decomposition*

$$\mathbb{F}_{2^n} = s \oplus c \oplus u \oplus w$$

with $\dim s = 10t, \dim c = 4$, $\dim u = \dim w = 10t - 2$. Each of the subspaces s, c, u, and w is invariant under σ. The restriction of the bilinear form $(\alpha, \beta) \mapsto \operatorname{Tr}_1^n(\alpha\beta)$ to $s \times s$ and to $c \times c$ is nondegenerate, its restrictions to $u \times u$ and $w \times w$ vanish, and the induced bilinear pairing between u and w is nondegenerate.

Proof. We first prove the dimensions from the primary decomposition and then use adjoint operators to determine which primary components are paired nondegenerately by the trace form. The factorization above and the Chinese remainder theorem give the four stated σ-invariant sums of primary components. The self-reciprocal factor has degree two and occurs with multiplicity two, so $\dim c = 4$. The kernel of Tr_m^n has dimension $2m = 20t$. The remaining $20t - 4$ dimensions are divided equally between reciprocal factors, which gives $\dim u = \dim w = 10t - 2$.

The adjoint of σ with respect to the bilinear form $(\alpha, \beta) \mapsto \operatorname{Tr}_1^n(\alpha\beta)$ on $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ is σ^{-1}. If $d_p = \deg p$, then $p^*(\sigma^{-1}) = \sigma^{-d_p} p(\sigma)$; hence the adjoint of $p^*(\sigma)^2$ is $\sigma^{-2d_p} p(\sigma)^2$. We take

irreducible factors $p, q \in \mathbb{F}_2[z]$ such that $q \neq p^*$. Since p^* and q are coprime, Bézout's identity shows that $p^*(\sigma)^2$ is invertible on the q^2-primary component. Hence every β in that component can be written as $\beta = p^*(\sigma)^2\beta'$ for some β' in the same q^2-primary component. If α belongs to the p^2-primary component, then we have

$$\mathrm{Tr}_1^n(\alpha\beta) = \mathrm{Tr}_1^n(\alpha p^*(\sigma)^2\beta') = \mathrm{Tr}_1^n(\sigma^{-2d_p}p(\sigma)^2\alpha\,\beta') = 0.$$

Thus the bilinear form vanishes between a primary component and every component except the one associated with its reciprocal factor. We suppose that α belongs to a p^2-primary component and that $\mathrm{Tr}_1^n(\alpha\beta) = 0$ for every β in the $(p^*)^2$-primary component. By the orthogonality just proved, the same equality holds for every β in every primary component of $\ker \mathrm{Tr}_m^n$; it also holds for every $\beta \in s$ because $v = s^\perp$. It therefore holds for all $\beta \in \mathbb{F}_{2^n}$, so $\alpha = 0$ by nondegeneracy of the bilinear form. Hence the restriction of this form to $u \times w$ is nondegenerate, whereas it vanishes on $u \times u$ and on $w \times w$. The same argument proves that the restriction to the self-reciprocal component c is nondegenerate. Finally, the restriction to s is the nondegenerate bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^m(\alpha\beta)$ on $s \times s$, because the extension degree is three, and the preceding trace identity gives $v = \ker \mathrm{Tr}_m^n = s^\perp$. This completes the proof. □

By the Chinese remainder theorem, the four groups of primary factors determine four pairwise orthogonal idempotents whose sum is one. Evaluating these idempotents at σ gives the $\mathbb{F}_2$-linear projections onto s, c, u, and w. Consequently, every $\alpha \in \mathbb{F}_{2^n}$ has a unique decomposition $\alpha = \alpha_s + \alpha_c + \alpha_u + \alpha_w$, where $\alpha_s \in s, \alpha_c \in c, \alpha_u \in u, \alpha_w \in w$. We refer to α_s, α_c, α_u, and α_w as the corresponding components of α. Since each projection is a polynomial in σ, all four projections commute with σ.

4.2 Bent functions on the subspace c with $\dim c = 4$

As an $\mathbb{F}_2[z]$-module, the component c is isomorphic to

$$c \cong \mathbb{F}_2[z]/\big((z^2+z+1)^2\big) = \mathbb{F}_2[z]/(z^4+z^2+1),$$

where the last equality follows from $(z^2+z+1)^2 = z^4+z^2+1$ over $\mathbb{F}_2$. In particular, $\dim_{\mathbb{F}_2} c = 4$. We write an element of c in the basis $1, z, z^2, z^3$ as $(c_0, c_1, c_2, c_3) \in \mathbb{F}_2^4$. Multiplication by z, and hence the restriction of σ to c, is given by $(c_0, c_1, c_2, c_3) \mapsto (c_3, c_0, c_1+c_3, c_2)$. We define $\kappa_0, \kappa_2 : c \to \mathbb{F}_2$ by $\kappa_0(c_0, c_1, c_2, c_3) = c_0c_1+c_1c_2+c_2c_3$ and $\kappa_2(c_0, c_1, c_2, c_3) = c_0+c_1+c_2+c_3+c_0c_2+c_1c_3$.

Proposition 4.2. *The functions κ_0 and κ_2 are invariant under the restriction of σ to c and are quadratic bent functions on c, and* $\deg(\kappa_0 + \kappa_2) = 2$.

Proof. Substitution of $(c_3, c_0, c_1 + c_3, c_2)$ for (c_0, c_1, c_2, c_3) leaves each function unchanged, so both functions are invariant under the restriction of σ to c.

For κ_0, the invertible linear change of variables $y_0 = c_1$, $y_1 = c_0 + c_2$, $y_2 = c_2$, and $y_3 = c_3$ transforms κ_0 into $y_0y_1 + y_2y_3$. For κ_2, the coordinate permutation $y_0 = c_0$, $y_1 = c_2$, $y_2 = c_1$, and $y_3 = c_3$ transforms κ_2 into $y_0y_1 + y_2y_3 + y_0 + y_1 + y_2 + y_3$. Since $y_0y_1 + y_2y_3$ is bent and the addition of an affine function preserves bentness, both κ_0 and κ_2 are quadratic bent functions. Their quadratic homogeneous parts are distinct, so $\deg(\kappa_0 + \kappa_2) = 2$. This finishes the proof. □

4.3 Bent functions on $s \oplus c$

For $s = \mathbb{F}_{2^{10t}}$, we use the function g and the functions g'_e defined in (1) and (3), respectively. For $\alpha_s \in s$ and $\alpha_c \in c$, we define $h_0 : s \oplus c \to \mathbb{F}_2$ by

$$h_0(\alpha_s, \alpha_c) = g(\alpha_s) + \kappa_0(\alpha_c).$$

We now construct a bent function h_e on $s \oplus c$ for every $1 \le e \le 5t+2$ such that $\deg(h_0 + h_e) = e$. For any $\alpha_s \in s$ and $\alpha_c \in c$, we define the functions $h_e : s \oplus c \to \mathbb{F}_2$ as follows. For $e = 1$, we put $h_1(\alpha_s, \alpha_c) = h_0(\alpha_s, \alpha_c) + \mathrm{Tr}_1^m(\alpha_s)$. For $e = 2$, we put $h_2(\alpha_s, \alpha_c) = g(\alpha_s) + \kappa_2(\alpha_c)$. For $3 \le e \le 5t$, we put $h_e(\alpha_s, \alpha_c) = g'_e(\alpha_s) + \kappa_0(\alpha_c)$. Finally, for $e \in \{5t+1, 5t+2\}$, we put $d = e - 2$ and define

$$h_e(\alpha_s, \alpha_c) = g(\alpha_s) + \kappa_0(\alpha_c) + \big(g(\alpha_s) + g'_d(\alpha_s)\big)\big(\kappa_0(\alpha_c) + \kappa_2(\alpha_c)\big).$$

Proposition 4.3. *For every $1 \le e \le 5t+2$, the function h_e above is invariant under the restriction of σ to $s \oplus c$ and is bent, and*

$$\deg(h_0 + h_e) = e.$$

Moreover,

$$\mathrm{ind}(h_0) \le 4t + 2. \tag{5}$$

Proof. For $1 \le e \le 5t$, bentness follows from an affine modification or a direct sum, whereas the two remaining cases follow from Carlet's indirect sum in Proposition 3.4. We now verify the corresponding degrees and invariance. The function h_1 is obtained from h_0 by adding the nonzero linear function $\mathrm{Tr}_1^m(\alpha_s)$. Hence it is bent, invariant under σ, and $\deg(h_0 + h_1) = 1$. The function h_2 is the direct sum of g and κ_2, so it is bent and invariant under σ, while Proposition 4.2 gives $\deg(h_0 + h_2) = 2$. For $3 \le e \le 5t$, the function h_e is the direct sum of g'_e and κ_0. Proposition 3.3 therefore gives its bentness, invariance, and $\deg(h_0 + h_e) = e$. For $e \in \{5t+1, 5t+2\}$, Proposition 3.4 applies to $g, g'_{e-2}, \kappa_0, \kappa_2$. It proves the bentness of h_e, while $h_0 + h_e = (g + g'_{e-2})(\kappa_0 + \kappa_2)$ has algebraic degree $(e-2) + 2 = e$. The functions g and g'_{e-2} are invariant under $\sigma|_s$, while κ_0 and κ_2 are invariant under $\sigma|_c$. Hence h_e is invariant under $\sigma|_{s \oplus c}$.

It remains to prove (5). We take an M-subspace $q \subseteq s \oplus c$ of h_0, and let $\pi_s : s \oplus c \to s$ be the coordinate projection. Since the second-order derivative of the quadratic function κ_0 is constant, $\pi_s(q)$ is a relaxed M-subspace of g and hence has dimension at most $4t$ by (2). Set $K = \ker(\pi_s|_q)$. Every vector in K has zero s-component. Consequently, for all $a, b \in K$, the identity $D_a D_b h_0 = 0$ reduces to $D_{a_c} D_{b_c} \kappa_0 = 0$. Under the invertible linear change of variables used in Proposition 4.2, κ_0 becomes $y_0 y_1 + y_2 y_3$. Define $J(y_0, y_1, y_2, y_3) = (y_1, y_0, y_3, y_2)$. A direct calculation gives $D_a D_b(y_0 y_1 + y_2 y_3) = a \cdot J(b)$, where $\cdot$ denotes the standard inner product on $\mathbb{F}_2^4$. Thus $J(K) \subseteq K^\perp$. Since J is invertible, $\dim J(K) = \dim K$, whereas $\dim K^\perp = 4 - \dim K$. It follows that $\dim K \le 2$. Therefore, $\dim q = \dim \pi_s(q) + \dim \ker(\pi_s|_q) \le 4t + 2$. This completes the proof.

□

4.4 Functions on u invariant under the Frobenius map

We index the irreducible factors in $\mathbb{F}_2[z]$ selected in the definition of u as $p_1, \dots, p_q$, where $q = 1 + 3j$, and we put $\ell_\nu = \deg p_\nu$. Their degrees form the multiset

$$\mathcal{L}_j = \{4\} \cup \bigcup_{i=1}^{j} \{6 \cdot 7^{i-1}, 12 \cdot 7^{i-1}, 12 \cdot 7^{i-1}\}.$$

The union is a multiset union. Its sum is

$$\sum_{\ell \in \mathcal{L}_j} \ell = 4 + 30 \sum_{i=1}^{j} 7^{i-1} = 5t - 1.$$

Let $u_\nu \subseteq u$ be the p_ν^2-primary component. Then u_ν is isomorphic, as an $\mathbb{F}_2[z]$-module, to $\mathbb{F}_2[z]/(p_\nu^2)$ and has dimension $2\ell_\nu$. Here $p_\nu(\sigma)u_\nu = \{p_\nu(\sigma)(\alpha) : \alpha \in u_\nu\}$. The quotient $u_\nu / p_\nu(\sigma)u_\nu$ is isomorphic to $\mathbb{F}_2[z]/(p_\nu)$ and therefore has dimension ℓ_ν. We let $\rho_\nu : u_\nu \to \mathbb{F}_2$ be the indicator of the nonzero elements of this quotient, composed with the quotient map. Since multiplication by z is invertible on the quotient, ρ_ν is invariant under σ. In binary coordinates it is the indicator of the nonzero vectors in an ℓ_ν-dimensional quotient, so its algebraic degree is exactly ℓ_ν.

We define $\mathcal{R}_j$ as the set consisting of all subset sums of $\mathcal{L}_j$, together with $10t - 2$. For $1 \le \nu \le q$, we let $\pi_\nu : u \to u_\nu$ be the coordinate projection. For every positive subset sum r, we choose a set $J_r \subseteq \{1, \dots, q\}$ such that $r = \sum_{\nu \in J_r} \ell_\nu$. For $\alpha_u \in u$, we define $\eta_r(\alpha_u) = \prod_{\nu \in J_r} \rho_\nu(\pi_\nu(\alpha_u))$. We define $\eta_0 : u \to \mathbb{F}_2$ as the zero function. For $r = 10t - 2$, we define $\eta_r : u \to \mathbb{F}_2$ as the indicator of $u \setminus \{0\}$. Each projection π_ν commutes with σ, and the factors in the defining product depend on disjoint primary components. Therefore every η_r is invariant under σ and has algebraic degree r.

Lemma 4.4. *Let $r_0 < r_1 < \cdots < r_h$ be the elements of $\mathcal{R}_j$. Then $r_0 = 0$, $r_1 = 4$, $r_h = 10t - 2$, and $r_{i+1} - r_i \le 5t - 1$ for all $0 \le i < h$. Consequently, we have $\{3\} \cup (\mathcal{R}_j \setminus \{0\}) \cup \{r + e : r \in \mathcal{R}_j \setminus \{0\},\ 1 \le e \le 5t + 2\} = \{3, 4, \dots, 15t\}$.*

Proof. The first positive entry is four by the definition of $\mathcal{L}_j$. To bound all subsequent gaps, we arrange the elements of the multiset as $\ell_1 \le \ell_2 \le \cdots \le \ell_q$ and let S_k be the set of subset sums of $\ell_1, \dots, \ell_k$. We prove by induction that the gap between two consecutive elements of S_k is at most ℓ_k. This is immediate for $k = 1$. Moreover, $S_k = S_{k-1} \cup (S_{k-1} + \ell_k)$. Gaps within either copy are at most $\ell_{k-1} \le \ell_k$. We take consecutive elements $x < y$ in the union, and we suppose that they belong to different copies. If $x \in S_{k-1}$ and $y \in S_{k-1} + \ell_k$, then either $x < \ell_k$, in which case $y = \ell_k$, or $y - \ell_k \le x$; in both cases $y - x \le \ell_k$. If $x \in S_{k-1} + \ell_k$ and $y \in S_{k-1}$, the preceding element of S_{k-1} before y cannot exceed x, and the induction hypothesis again gives $y - x \le \ell_k$. This proves the induction claim.

The largest entry of $\mathcal{L}_j$ is at most $12 \cdot 7^{j-1}$ when $j \ge 1$, and is four when $j = 0$. Both bounds are at most $5t - 1$. Since the sum of all elements of $\mathcal{L}_j$ is $5t - 1$, the largest subset sum is $5t - 1$. The additional entry $10t - 2$ is twice this value, so the final gap is also $5t - 1$.

For $1 \le i \le h$, we define $I_i = \{r_i + 1, r_i + 2, \dots, r_i + 5t + 2\}$. Since $r_{i+1} - r_i \le 5t - 1$, the end of I_i is at least the beginning of I_{i+1}; thus consecutive intervals overlap. The first interval begins at five because $r_1 = 4$, and the last ends at $(10t - 2) + (5t + 2) = 15t$. The separate degrees three and four complete the stated interval. This completes the proof. □

4.5 Functions generated by the first construction

For $s \oplus c$, we use the function h_0 and the functions h_e defined above. Set $h'_e = h_e$ for $1 \le e \le 5t+2$, and additionally define $h'_0 : s \oplus c \to \mathbb{F}_2$ by $h'_0 = h_0 + 1$. Then h'_0 is bent and invariant under the restriction of σ to $s \oplus c$, and $h_0 + h'_0 = 1$. For $r \in \mathcal{R}_j$ and $e = 0$, or for $r \in \mathcal{R}_j \setminus \{0\}$ and $1 \le e \le 5t+2$, define $f_{r,e} : \mathbb{F}_{2^n} \to \mathbb{F}_2$ by

$$f_{r,e}(\alpha) = \mathrm{Tr}_1^n(\alpha_u \alpha_w) + h_0(\alpha_s, \alpha_c) + \eta_r(\alpha_u)(h_0(\alpha_s, \alpha_c) + h'_e(\alpha_s, \alpha_c)), \text{ where } \alpha \in \mathbb{F}_{2^n}. \quad (6)$$

For $e = 0$, we write $f_r = f_{r,0}$. Since $h_0 + h'_0 = 1$, we have $f_r(\alpha) = \mathrm{Tr}_1^n(\alpha_u \alpha_w) + h_0(\alpha_s, \alpha_c) + \eta_r(\alpha_u)$.

Theorem 4.5. *For every $j \ge 0$, put $n = 30 \cdot 7^j$. For every integer d with $3 \le d \le n/2$, the functions defined in (6) contains an idempotent bent function $f_{n,d} : \mathbb{F}_{2^n} \to \mathbb{F}_2$ of algebraic degree d outside $\mathcal{M}^\#$.*

Proof. We verify Frobenius invariance, bentness, algebraic degree, and the fact that the functions lie outside $\mathcal{M}^\#$.

All four projections commute with σ, the trace function Tr_1^n is unchanged by squaring, and every function used on s, c, and u is invariant under the corresponding restriction of σ. Since $h'_0 = h_0 + 1$ and $h'_e = h_e$ for $1 \le e \le 5t+2$, every h'_e is also invariant under σ. Hence every function in (6) is invariant under $\alpha \mapsto \alpha^2$. Proposition 4.1 gives a nondegenerate bilinear form between u and w. We apply Theorem 3.1 with $V_0 = s \oplus c$, $U = u$, $W = w$, $k = 10t - 2$, and $b(\beta, \gamma) = \mathrm{Tr}_1^n(\beta\gamma)$. If $e = 0$, then $h_0 + h'_0 = 1$, so $f_{r,0}(\alpha) = \mathrm{Tr}_1^n(\alpha_u \alpha_w) + h_0(\alpha_s, \alpha_c) + \eta_r(\alpha_u)$ and $f_{r,0}$ is the function h_{η_r} in Theorem 3.1. If $1 \le e \le 5t+2$, then $h'_e = h_e$ and $f_{r,e}$ is the function h_{η_r, h_e} in that theorem. Proposition 4.3 shows that h_0 and h_e are bent, so every function in (6) is bent.

For $e = 0$ and $r = 0$, the function $f_{0,0}$ has algebraic degree three. For $e = 0$ and $r \in \mathcal{R}_j \setminus \{0\}$, it has algebraic degree r, since η_r has degree $r \ge 4$, whereas $\mathrm{Tr}_1^n(\alpha_u \alpha_w)$ and h_0 have degrees two and three, respectively. For $1 \le e \le 5t+2$, Proposition 4.3 gives $\deg(h_0 + h_e) = e$. The two factors in the last term of (6) depend on the complementary subspaces u and $s \oplus c$, so their nonzero homogeneous parts of degrees r and e have a nonzero product. Consequently, $\deg f_{r,e} = r + e$. Lemma 4.4 therefore shows that every algebraic degree from three to $15t = n/2$ occurs.

It remains to prove that these functions lie outside $\mathcal{M}^\#$. If $e = 0$, the first index bound in Theorem 3.1, together with (5), gives $\mathrm{ind}(f_{r,0}) \le (10t - 2) + (4t + 2) = 14t < 15t = n/2$. If $1 \le e \le 5t+2$, then $r > 0$, so η_r is nonconstant and the second index bound in Theorem 3.1 gives the same estimate, namely $\mathrm{ind}(f_{r,e}) \le 14t < n/2$. Thus no function in (6) has an $(n/2)$-dimensional M-subspace, and the criterion in Section 2 proves that these functions do not belong to $\mathcal{M}^\#$. This completes the proof. □

Example 4.6. Let $j = 0$. Then $n = 30$, $\mathcal{L}_0 = \{4\}$, and $\mathcal{R}_0 = \{0, 4, 8\}$. The function f_0 has algebraic degree three, while f_4 has algebraic degree four. Moreover, the functions $f_{4,e}$ have algebraic degrees $5, \ldots, 11$, and the functions $f_{8,e}$ have algebraic degrees $9, \ldots, 15$. Hence this construction produces 30-variable bent functions of every algebraic degree from 3 to 15. The bentness and algebraic degrees follow from the verified properties of the 10-variable function, Propositions 3.4 and 4.3,and values of the Walsh transform follow from the proof of Theorem 3.1.

5 Rotation-Symmetric Bent Functions Outside $\mathcal{M}^{\#}$ with Any Possible Algebraic Degree for $n = 70t$ Variables

The construction in Section 4 applies to $n = 30 \cdot 7^j$ and involves a four-dimensional component associated with a self-reciprocal factor. In this section, for every integer $t \geq 1$ and every d satisfying $3 \leq d \leq n/2$, we construct a rotation-symmetric bent function of algebraic degree d outside the completed Maiorana–McFarland class, where $n = 70t$. We put $m = 10t$ and view $\mathbb{F}_{2^n}$ as an extension of degree seven of $\mathbb{F}_{2^m}$. The kernel of the relative trace map Tr_m^n is decomposed into two Frobenius-invariant subspaces associated with the two reciprocal cubic factors of $z^7 - 1$. The map $(\alpha, \beta) \mapsto \mathrm{Tr}_1^n(\alpha\beta)$ defines a nondegenerate bilinear pairing between these two subspaces, which will be used in the construction below.

5.1 Decomposition of $\mathbb{F}_{2^n}$ over $\mathbb{F}_{2^m}$

Fix an integer $t \geq 1$, put $m = 10t$ and $n = 7m = 70t$, and regard $\mathbb{F}_{2^n}$ as an extension of degree seven of $s = \mathbb{F}_{2^m}$. Let $\sigma : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ be the Frobenius map defined by $\sigma(\alpha) = \alpha^2$, and set $\tau = \sigma^m$. The relative trace map is $\ell_0 = \mathrm{Tr}_m^n = 1 + \tau + \cdots + \tau^6$. Since τ fixes every element of s and the extension degree is odd, the restriction of ℓ_0 to s is the identity. Thus ℓ_0 is a projection onto s, and $\mathbb{F}_{2^n} = s \oplus \ker \ell_0$. Moreover, for every $\alpha \in \mathbb{F}_{2^n}$ and $\beta \in s$, the trace identity $\mathrm{Tr}_1^n(\alpha\beta) = \mathrm{Tr}_1^m(\beta\ell_0(\alpha))$ shows that $\ker \ell_0 = s^{\perp}$. Hence the two subspaces in this decomposition are orthogonal with respect to the bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^n(\alpha\beta)$.

In $\mathbb{F}_2[z]/(z^7 - 1)$, we define $e(z) = 1 + z + z^2 + z^4$ and $e^{\dagger}(z) = 1 + z^3 + z^5 + z^6$. The two idempotents correspond to the reciprocal factors $z^3 + z + 1$ and $z^3 + z^2 + 1$, respectively. Their linearized maps on $\mathbb{F}_{2^n}$ are $\ell, \ell^{\dagger} : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$. For $\alpha \in \mathbb{F}_{2^n}$, they are given by $\ell(\alpha) = \alpha + \alpha^{2^m} + \alpha^{2^{2m}} + \alpha^{2^{4m}}$ and $\ell^{\dagger}(\alpha) = \alpha + \alpha^{2^{3m}} + \alpha^{2^{5m}} + \alpha^{2^{6m}}$.

Proposition 5.1. *Let $u = \ell(\mathbb{F}_{2^n})$ and $w = \ell^{\dagger}(\mathbb{F}_{2^n})$. Then we have*

$$\mathbb{F}_{2^n} = s \oplus u \oplus w,$$

where $\dim s = m$ and $\dim u = \dim w = 3m$. All three spaces are invariant under σ. Moreover, the bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^n(\alpha\beta)$ is nondegenerate on $u \times w$ and vanishes on $u \times u$ and on $w \times w$.

Proof. We first use the three idempotents to identify the three subspaces in the direct-sum decomposition and determine their dimensions. We then use the adjoint relation between e and $e^{\dagger}$ to prove the assertions concerning the trace form. Multiplication modulo $z^7 - 1$ gives $e^2 = e$, $(e^{\dagger})^2 = e^{\dagger}$, and $ee^{\dagger} = 0$. If $q(z) = 1 + z + \cdots + z^6$, then $q^2 = q$, $qe = qe^{\dagger} = 0$, and $e + e^{\dagger} = 1 + q$. The map $q(\tau) = \ell_0$ has image s and restricts to the identity on s. Hence it is the projection onto s along $\ker \ell_0$. Since subtraction and addition coincide over $\mathbb{F}_2$, the complementary map $I + q(\tau)$ is the projection onto $\ker \ell_0$ along s. It follows that $u, w \subseteq \ker \ell_0$ and $u \oplus w = \ker \ell_0$. A normal basis of $\mathbb{F}_{2^n}/\mathbb{F}_{2^m}$ represents the action of τ by cyclic shift on seven coordinates over $\mathbb{F}_{2^m}$. Evaluation at a root of $z^3 + z + 1$ shows that e is one on the component obtained from the Chinese remainder theorem for this factor and zero on the other two; hence u is the component for $z^3 + z + 1$. Similarly, w is the component for $z^3 + z^2 + 1$. Both factors have degree three, so u and w each have dimension three over $\mathbb{F}_{2^m}$.

The adjoint of τ^i with respect to the bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^n(\alpha\beta)$ is τ^{-i}. Replacing z by z^{-1} sends e to $e^{\dagger}$. Therefore, for all $\alpha, \beta \in \mathbb{F}_{2^n}$, we have

$$\mathrm{Tr}_1^n(\ell(\alpha)\ell(\beta)) = \mathrm{Tr}_1^n(\alpha\ell^{\dagger}\ell(\beta)) = 0.$$

The corresponding equality on w follows by interchanging ℓ and $\ell^\dagger$. We have $\ker \ell_0 = s^\perp$ because, for $\alpha \in \mathbb{F}_{2^n}$ and $\beta \in s$, $\mathrm{Tr}_1^n(\alpha\beta) = \mathrm{Tr}_1^m(\beta\ell_0(\alpha))$. The restriction of the bilinear form to s is nondegenerate and therefore its restriction to $s^\perp$ is also nondegenerate. We suppose that $\alpha \in u$ satisfies $\mathrm{Tr}_1^n(\alpha\beta) = 0$ for every $\beta \in w$. The same equality holds for every $\beta \in u$ because the bilinear form vanishes on $u \times u$, and it holds for every $\beta \in s$ because $u \subseteq s^\perp$. Hence it holds for every $\beta \in \mathbb{F}_{2^n}$, so $\alpha = 0$. The same argument with u, w interchanged proves that the bilinear form is nondegenerate on $u \times w$. Finally, σ commutes with τ, ℓ, and $\ell^\dagger$, so each subspace in this direct-sum decomposition is invariant under σ. This completes the proof. $\square$

In $\mathbb{F}_2[z]/(z^7 - 1)$, the three polynomials $q, e, e^\dagger$ are pairwise orthogonal idempotents whose sum is one. Therefore, ℓ_0, ℓ, and $\ell^\dagger$ are complementary idempotent projections onto s, u, and w, respectively, rather than merely maps with images contained in these subspaces. For $\alpha \in \mathbb{F}_{2^n}$, they give

$$\alpha_s = \ell_0(\alpha), \quad \alpha_u = \ell(\alpha), \quad \alpha_w = \ell^\dagger(\alpha), \quad \alpha = \alpha_s + \alpha_u + \alpha_w. \tag{7}$$

5.2 Functions on u invariant under the Frobenius map

For an $\mathbb{F}_8$-valued function φ on u, we define the action of σ by $(\sigma\varphi)(\alpha) = \varphi(\sigma^{-1}\alpha)$ for every $\alpha \in u$, and we define $\sigma^i\varphi$ by iteration.

Lemma 5.2. *For every $d \in \{3, 4, \ldots, 3m-3, 3m\}$, there exist d distinct linear forms $\lambda_{d,1}, \ldots, \lambda_{d,d}$ selected from a τ-eigenbasis of $u^* \otimes_{\mathbb{F}_2} \mathbb{F}_8$ and an element $\gamma_d \in \mathbb{F}_8$ such that, with $p_d(\alpha) = \prod_{j=1}^d \lambda_{d,j}(\alpha)$, the function*

$$\eta_d(\alpha) = \mathrm{Tr}_1^3\left(\gamma_d \sum_{\varphi \in \mathcal{O}_\sigma(p_d)} \varphi(\alpha)\right), \quad \alpha \in u, \tag{8}$$

is a Boolean function invariant under σ and has algebraic degree d, where $\mathcal{O}_\sigma(p_d) = \{\sigma^i p_d : i \geq 0\}$ denotes the set of distinct σ-translates of p_d under the action defined above.

Proof. We first construct the required products for $3 \leq d \leq 3m/2$, then obtain the degrees $3m/2 < d \leq 3m - 3$ by taking complementary products, and finally treat $d = 3m$.

We write $\mathbb{F}_8 = \mathbb{F}_2(\omega)$, where $\omega^3 + \omega + 1 = 0$. On $u \otimes_{\mathbb{F}_2} \mathbb{F}_8$, the three eigenvalues of τ are ω, ω^2, and ω^4, each with multiplicity m. For $\lambda \in u^*$ and $\alpha \in u$, we let τ act on the dual by $(\tau\lambda)(\alpha) = \lambda(\tau^{-1}\alpha)$ and extend this action to $u^* \otimes_{\mathbb{F}_2} \mathbb{F}_8$. Its eigenvalues are therefore ω^6, ω^5, and ω^3. We label the corresponding eigenspaces as types 1, 2, and 4, respectively, so that a type-a form has eigenvalue ω^{-a}.

We briefly explain how algebraic degree behaves after extending scalars. Let $\mathcal{B}_{\leq d}(u)$ be the space of Boolean functions on u of algebraic degree at most d. Every invertible linear map preserves the filtration $\mathcal{B}_{\leq 0}(u) \subseteq \mathcal{B}_{\leq 1}(u) \subseteq \cdots \subseteq \mathcal{B}_{\leq 3m}(u)$ and induces an invertible action on $\mathcal{B}_{\leq d}(u)/\mathcal{B}_{\leq d-1}(u)$. After fixing a coordinate basis, this quotient has a basis consisting of the classes of products of d distinct coordinate forms. We extend this quotient from $\mathbb{F}_2$ to $\mathbb{F}_8$ and choose a τ-eigenbasis of $u^* \otimes_{\mathbb{F}_2} \mathbb{F}_8$ consisting of m forms of each type. A product containing c_1, c_2, and c_4 forms of types 1, 2, and 4, respectively, has eigenvalue $\omega^{-(c_1+2c_2+4c_4)}$. It is fixed by τ precisely when $c_1 + 2c_2 + 4c_4 \equiv 0 \pmod 7$.

We next choose the eigenbasis compatibly with the action of σ. With respect to a normal basis and after extending scalars to $\mathbb{F}_8$, the cyclic module for σ is $\mathbb{F}_8[z]/(z^{7m} - 1)$.

For $a \in \{1,2,4\}$, the factor $z^m - \omega^{-a}$ corresponds to the type-a eigenspace. The Chinese remainder theorem identifies this eigenspace with $\mathbb{F}_8[z]/(z^m - \omega^{-a})$. We denote the linear forms corresponding to $1, z, \ldots, z^{m-1}$ by $x_{0,a}, \ldots, x_{m-1,a}$. Thus σ advances the index and introduces the factor ω^{-a} when the index passes from $m-1$ to 0.

Let $3 \le d \le 3m/2$ and write $d = 3q + \epsilon$, where $\epsilon \in \{0,1,2\}$. We choose

$$(c_1, c_2, c_4) = \begin{cases} (q, q, q), & \epsilon = 0, \\ (q+2, q-1, q), & \epsilon = 1, \\ (q+2, q+1, q-1), & \epsilon = 2. \end{cases}$$

Each displayed triple has sum d and weighted sum $c_1 + 2c_2 + 4c_4 = 7q$. The boundary values $d = 3, 4, 5$ give $(1,1,1)$, $(3,0,1)$, and $(3,2,0)$, respectively, while $d = 3m/2$ gives $(m/2, m/2, m/2)$. Since $m \ge 10$ and $d \le 3m/2$, every entry lies between zero and $m-1$. For each $a \in \{1,2,4\}$, set $I_{a,d} = \{0, 1, \ldots, c_a - 1\}$, with $I_{a,d} = \varnothing$ when $c_a = 0$, and define $p_d(\alpha) = \prod_{a\in\{1,2,4\}} \prod_{i\in I_{a,d}} x_{i,a}(\alpha)$. The factors in this product are d distinct members of the chosen τ-eigenbasis. At least one $I_{a,d}$ is nonempty, and each nonempty $I_{a,d}$ is a proper interval of cyclically indexed positions. Hence only the identity cyclic shift fixes all three sets. Moreover, the weighted congruence gives $\sigma^m(p_d) = p_d$. Consequently, $\mathcal{O}_\sigma(p_d) = \{p_d, \sigma(p_d), \ldots, \sigma^{m-1}(p_d)\}$, and their terms of algebraic degree d correspond to nonzero scalar multiples of distinct basis monomials.

For $3m/2 < d \le 3m - 3$, put $d' = 3m - d$ and construct $I_{a,d'}$ as above. We then set $I_{a,d} = \{0, 1, \ldots, m-1\} \setminus I_{a,d'}$ and define p_d by the same product. The weighted sum of all $3m$ eigenforms is $m(1+2+4) = 7m$, so the complemented product also satisfies the required congruence and has algebraic degree d. A cyclic shift fixes a set if and only if it fixes its complement. Thus the orbit of p_d again has m elements whose terms of algebraic degree d are nonzero scalar multiples of distinct basis monomials.

Finally, for $d = 3m$, define $p_{3m}(\alpha) = \prod_{a\in\{1,2,4\}} \prod_{i=0}^{m-1} x_{i,a}(\alpha)$. This is the product of all $3m$ members of the chosen eigenbasis and therefore has algebraic degree $3m$. The action of σ introduces the total scalar factor $\omega^{-(1+2+4)} = \omega^{-7} = 1$, so $\sigma(p_{3m}) = p_{3m}$ and $\mathcal{O}_\sigma(p_{3m}) = \{p_{3m}\}$.

For every $d \in \{3, 4, \ldots, 3m-3, 3m\}$, set $q_d(\alpha) = \sum_{\varphi\in\mathcal{O}_\sigma(p_d)} \varphi(\alpha)$. The action of σ permutes the elements of $\mathcal{O}_\sigma(p_d)$, so q_d is invariant under σ. For $d \le 3m-3$, its distinct orbit terms cannot cancel in algebraic degree d, while for $d = 3m$ we have $q_{3m} = p_{3m}$. Hence $\deg q_d = d$ in every case.

Fix a binary coordinate basis of u^* and expand q_d in the ANF monomial basis over $\mathbb{F}_8$. Let $\theta_d \in \mathbb{F}_8^*$ be a nonzero coefficient of algebraic degree d. Since the bilinear form $(\alpha, \beta) \mapsto \mathrm{Tr}_1^3(\alpha\beta)$ on $\mathbb{F}_8 \times \mathbb{F}_8$ is nondegenerate, there exists $\gamma_d \in \mathbb{F}_8$ such that $\mathrm{Tr}_1^3(\gamma_d\theta_d) = 1$. The function η_d in (8) is therefore Boolean and retains a nonzero coefficient of algebraic degree d, so $\deg \eta_d = d$. Since q_d is invariant under σ, we also have $\eta_d(\alpha^2) = \eta_d(\alpha)$ for every $\alpha \in u$. This completes the proof. □

5.3 Functions generated by the second construction

We set $d_m = \{0, 3, 4, \ldots, 3m-3, 3m\}$. For every positive $r \in d_m$, we choose one of the functions provided by Lemma 5.2 with $d = r$ and denote it by η_r. Thus η_r is invariant under σ and has algebraic degree r. We also define $\eta_0(\alpha) = 0$ for every $\alpha \in u$. On $s = \mathbb{F}_{2^{10t}}$, we use g and g'_e defined in (1) and (3), respectively. We additionally define $g'_0 : s \to \mathbb{F}_2$ by $g'_0 = g + 1$. The function g'_0 is bent and invariant under σ, and $g + g'_0 = 1$. For $r \in d_m$ and $e = 0$, or for

$r \in d_m \setminus \{0\}$ and $3 \le e \le 5t$, we define $h_{r,e} : \mathbb{F}_{2^n} \to \mathbb{F}_2$ by

$$h_{r,e}(\alpha) = \mathrm{Tr}_1^n(\alpha_u \alpha_w) + g(\alpha_s) + \eta_r(\alpha_u)(g(\alpha_s) + g'_e(\alpha_s)), \quad \alpha \in \mathbb{F}_{2^n}. \tag{9}$$

For $e = 0$, we write $h_r = h_{r,0}$. Since $g + g'_0 = 1$, we have $h_r(\alpha) = \mathrm{Tr}_1^n(\alpha_u \alpha_w) + g(\alpha_s) + \eta_r(\alpha_u)$.

Lemma 5.3. *For $m = 10t$, we have*

$$(d_m \setminus \{0\}) \cup \{r + e : r \in d_m \setminus \{0\},\ 3 \le e \le 5t,\ r + e \le 35t\} = \{3, 4, \ldots, 35t\}.$$

Proof. The set $d_m \setminus \{0\}$ contains $3, 4, \ldots, 3m-3$ and $3m$, so $3m$ itself is already covered by h_{3m}. The four integers $3m-2, 3m-1, 3m+1$, and $3m+2$ have representations $3m-2 = (3m-5)+3$, $3m-1 = (3m-4)+3$, $3m+1 = (3m-3)+4$ and $3m+2 = (3m-3)+5$, respectively. Every integer from $3m+3$ through $7m/2 = 35t$ is of the form $3m+e$ for some $3 \le e \le m/2 = 5t$. In each of the above representations, the integer is written as $r+e$, where $r \in d_m \setminus \{0\}$ and $3 \le e \le 5t$. Hence all these integers belong to the set on the left-hand side of the asserted equality. This completes the proof. □

Theorem 5.4. *Let $t \ge 1$ and $n = 7m = 70t$. Every function $h_{r,e} \in \mathcal{B}_n$ defined in (9), where either $r \in d_m = \{0, 3, 4, \ldots, 3m-3, 3m\}$ and $e = 0$, or $r \in d_m \setminus \{0\}$ and $3 \le e \le 5t$, is an idempotent bent function outside $\mathcal{M}^\#$. Its algebraic degree is*

$$\deg h_{r,e} = \begin{cases} 3, & (r,e) = (0,0), \\ r, & r > 0 \text{ and } e = 0, \\ r + e, & r > 0 \text{ and } 3 \le e \le 5t. \end{cases}$$

Consequently, the family in (9) contains an idempotent bent function outside $\mathcal{M}^\#$ of every algebraic degree from 3 to $n/2$.

Proof. The projections in (7) commute with σ, and all functions on s and u are invariant under the corresponding restrictions of σ. Thus both formulas are invariant under σ.

Proposition 5.1 gives a nondegenerate bilinear form between the $30t$-dimensional spaces u and w. We apply Theorem 3.1 with $V_0 = s$, $U = u$, $W = w$, $k = 30t$, $b(\beta, \gamma) = \mathrm{Tr}_1^n(\beta\gamma)$ for $\beta \in u$ and $\gamma \in w$, and $g_0 = g$. This proves that h_r is bent. For $h_{r,e}$, we take $g_1 = g'_e$; Propositions 3.2 and 3.3 show that both g and g'_e are bent.

The function h_0 has algebraic degree three. If $r = 3$, the cubic parts of $g(\alpha_s)$ and $\eta_3(\alpha_u)$ use disjoint variable sets and contain different monomials, so they cannot cancel. If $r > 3$, the homogeneous part of algebraic degree r of η_r cannot be cancelled by the quadratic function $(\alpha_u, \alpha_w) \mapsto \mathrm{Tr}_1^n(\alpha_u \alpha_w)$ or the cubic function g. Thus $\deg h_r = r$ for every positive $r \in d_m$. Moreover, (4) and (9) give $\deg h_{r,e} = r + e$: the two nonzero homogeneous factors of algebraic degrees r and e use disjoint variable sets, and the remaining terms have degree at most three. Lemma 5.3 supplies every algebraic degree from three to $35t = n/2$.

Finally, $\mathrm{ind}(g) \le \text{r-ind}(g)$. For h_r, the first index bound in Theorem 3.1 applies for every $r \in d_m$, including $r = 0$. For $h_{r,e}$ with $e \ge 3$, the definition requires $r > 0$; hence η_r has positive algebraic degree and is nonconstant, so the second index bound in Theorem 3.1 applies. In both cases, Theorem 3.1 and (2) bound every M-subspace by

$$30t + \mathrm{ind}(g) \le 30t + 4t = 34t < 35t = n/2.$$

Thus none of these functions belongs to $\mathcal{M}^\#$. This completes the proof. □

Remark 5.5. For $n = 70$, the functions $h_{r,0}$ give algebraic degrees $3, \ldots, 27$ and 30. Algebraic degrees $28, 29, 31, 32, 33, 34, 35$ are given respectively by $h_{25,3}, h_{26,3}, h_{27,4}, h_{27,5}, h_{30,3}, h_{30,4}, h_{30,5}$. Thus the smallest case of the second construction can also be checked directly from the stated degree sets.

6 Comparison with Existing Constructions

In this section, we compare the algebraic degree ranges established above with those of earlier constructions and determine whether the functions under comparison belong to $\mathcal{M}^{\#}$. Rotation-symmetric Boolean functions and finite field idempotents correspond under normal bases [2, 23]. Su and Tang constructed rotation-symmetric bent functions of every possible algebraic degree by modifying a quadratic bent function on a union of affine subspaces [22]. Tang et al. subsequently gave a finite field construction of bent idempotents with the corresponding algebraic degrees [23]. The following two propositions prove that the functions in these constructions belong to $\mathcal{M}^{\#}$.

Proposition 6.1. *Let r be a positive integer, and let $\Gamma \subseteq \mathbb{F}_2^r$ be closed under cyclic shifts. The Su–Tang construction modifies the quadratic bent function $(x, y) \mapsto x \cdot y$ on*

$$T_\Gamma = \bigcup_{\gamma \in \Gamma} \{(x, y) \in \mathbb{F}_2^r \times \mathbb{F}_2^r : y = x + \gamma\}.$$

The resulting rotation-symmetric bent function

$$f_\Gamma(x, y) = x \cdot y + \mathbf{1}_{T_\Gamma}(x, y) = x \cdot y + \mathbf{1}_\Gamma(x + y)$$

belongs to $\mathcal{M}^{\#}$.

Proof. We use the invertible linear change of variables $u = x$ and $v = x + y$. Then $y = u + v$ and therfore

$$f_\Gamma(u, u + v) = u \cdot v + \mathbf{1}_\Gamma(v) + \sum_{i=1}^{r} u_i.$$

The function $(u, v) \mapsto u \cdot v + \mathbf{1}_\Gamma(v)$ belongs to the Maiorana–McFarland class with the identity permutation, and $\sum_{i=1}^{r} u_i$ is affine. Hence f_Γ is extended-affine equivalent to a function in $\mathcal{M}$, and therefore $f_\Gamma \in \mathcal{M}^{\#}$. □

Proposition 6.2. *Let r be a positive integer, and let θ be a normal element of $\mathbb{F}_{2^r}$ over $\mathbb{F}_2$. Let*

$$g(x) = \sum_{i=0}^{r-1} c_i \operatorname{Tr}_1^{2r}\left(x^{2^i+1}\right) + c_r \operatorname{Tr}_1^{r}\left(x^{2^r+1}\right) + \epsilon, \quad \text{where } x \in \mathbb{F}_{2^{2r}},$$

be a quadratic bent idempotent, where $c_0, \ldots, c_r, \epsilon \in \mathbb{F}_2$. For $\phi \in \mathcal{B}_r$, define $f_{\phi,\theta} : \mathbb{F}_{2^{2r}} \to \mathbb{F}_2$ by

$$f_{\phi,\theta}(x) = g(x) + \phi(L_0(x), \ldots, L_{r-1}(x)),$$

where $L_i(x) = \operatorname{Tr}_1^{2r}(\theta^{2^i} x)$. Then $f_{\phi,\theta}$ is a bent function from the construction of Tang et al. [23, Theorem 4.7] and belongs to $\mathcal{M}^{\#}$. If ϕ is rotation-symmetric, then $f_{\phi,\theta}$ is a bent idempotent from [23, Corollary 4.8] and still belongs to $\mathcal{M}^{\#}$.

Proof. We first determine the common kernel of $L_0, \ldots, L_{r-1}$. For every $a \in \mathbb{F}_{2^r}$, it can be easily seen that $L_i(a) = \mathrm{Tr}_1^r(\mathrm{Tr}_r^{2r}(\theta^{2^i} a)) = \mathrm{Tr}_1^r(\theta^{2^i}\,\mathrm{Tr}_r^{2r}(a)) = \mathrm{Tr}_1^r(\theta^{2^i}(a + a^{2^r})) = 0$. Hence $\mathbb{F}_{2^r}$ is contained in the common kernel. If $\lambda_0, \ldots, \lambda_{r-1} \in \mathbb{F}_2$ satisfy $\sum_{i=0}^{r-1} \lambda_i L_i = 0$, then

$$\mathrm{Tr}_1^{2r}\left(\left(\sum_{i=0}^{r-1} \lambda_i \theta^{2^i}\right) x\right) = 0$$

for every $x \in \mathbb{F}_{2^{2r}}$ and therefore $\sum_{i=0}^{r-1} \lambda_i \theta^{2^i} = 0$. Note that $\theta, \theta^2, \ldots, \theta^{2^{r-1}}$ form a basis of $\mathbb{F}_{2^r}$ over $\mathbb{F}_2$. Thus all the coefficients λ_i's are zero. So we have the linear forms $L_0, \ldots, L_{r-1}$ are linearly independent. Thus the common kernel has dimension $2r - r = r$ and is therefore exactly $\mathbb{F}_{2^r}$.

We next show that $\mathbb{F}_{2^r}$ is an M-subspace of $f_{\phi,\theta}$. Since g is quadratic, its second-order derivatives are constant. For any $a, b \in \mathbb{F}_{2^r}$, we have

$$D_a D_b g = \sum_{i=0}^{r-1} c_i\,\mathrm{Tr}_1^{2r}(a^{2^i} b + a b^{2^i}) + c_r\,\mathrm{Tr}_1^r(a^{2^r} b + a b^{2^r}).$$

It follows from $a^{2^i} b + a b^{2^i} \in \mathbb{F}_{2^r}$ that $\mathrm{Tr}_1^{2r}(a^{2^i} b + a b^{2^i}) = 0$. In addition, $\mathrm{Tr}_1^r(a^{2^r} b + a b^{2^r}) = \mathrm{Tr}_1^r(ab + ab) = 0$ for any $a, b \in \mathbb{F}_{2^r}$. Hence $D_a D_b g = 0$. Moreover, recall that, for any $0 \le i \le r-1$, we have $L_i(a) = 0$ for every $a \in \mathbb{F}_{2^r}$. So we have

$$\phi(L_0(x + a), \ldots, L_{r-1}(x + a)) = \phi(L_0(x), \ldots, L_{r-1}(x)).$$

So we have $D_a D_b f_{\phi,\theta} = 0$ for all $a, b \in \mathbb{F}_{2^r}$.

From what has been discussed above, $\mathbb{F}_{2^r}$ is an r-dimensional M-subspace of the bent function $f_{\phi,\theta}$. The criterion in Section 2 now gives $f_{\phi,\theta} \in \mathcal{M}^{\#}$. This completes the proof. □

For $n = 2r$, the degree of a bent function is at most r, and Lemma 2.2 shows that algebraic degree two cannot occur outside $\mathcal{M}^{\#}$. The complete relevant interval is therefore $3, 4, \ldots, r$. Propositions 6.1 and 6.2 show that the constructions given in [23] and [22] do not provide bent functions outside $\mathcal{M}^{\#}$. The rotation-symmetric functions studied in [16] include a family of algebraic degree $n/2$ outside $\mathcal{M}^{\#}$ for every even $n \ge 8$, a quartic family for infinitely many values of n, and cubic examples in 10 variables. To the best of our knowledge, our two constructions are the first infinite class of rotation-symmetric bent functions in which functions have algebraic degrees ranging from 3 to $n/2$ while remaining entirely outside the completed Maiorana–McFarland class.

7 Conclusion

In this paper, we presented two constructions of infinite families of rotation-symmetric bent functions outside $\mathcal{M}^{\#}$. The first construction applies to $n = 30 \cdot 7^j$ for every $j \ge 0$. The self-reciprocal factor $z^2 + z + 1$ gives a subspace c with $\dim c = 4$, and each remaining component of the kernel of Tr_m^n is associated with one member of a reciprocal pair of irreducible factors. The second construction applies to $n = 70t$ for every $t \ge 1$ and uses the two reciprocal cubic factors of $z^7 - 1$.

For both constructions, the Walsh-transform calculation in Theorem 3.1 proves bentness. The upper bounds $14 \cdot 7^j < 15 \cdot 7^j$ and $34t < 35t$ for the dimensions of M-subspaces prove that

the resulting functions lie outside $\mathcal{M}^{\#}$. We also proved that the functions generated by either construction attain any possible algebraic degree ranging from 3 to $n/2$ for every value of n in the corresponding parameter set. By Lemma 2.2, this is the complete possible degree range outside $\mathcal{M}^{\#}$. To the best of our knowledge, these are the first infinite constructions that, for each stated value of n, attain any possible algebraic degree ranging from 3 to $n/2$ while all the resulting functions remain outside $\mathcal{M}^{\#}$.

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